A gene expression signature for survival and risk prediction of Breast Cancer that improves the signatures used in clinical genomic platforms.

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Background

Several gene signatures to assign risk to breast cancer have been implemented in the clinic. However, these are black boxes where the influence of markers is unclear and the risk scores have no clear biological interpretation. In addition, they can be unrelated to standard tumor markers that guide clinical decisions.

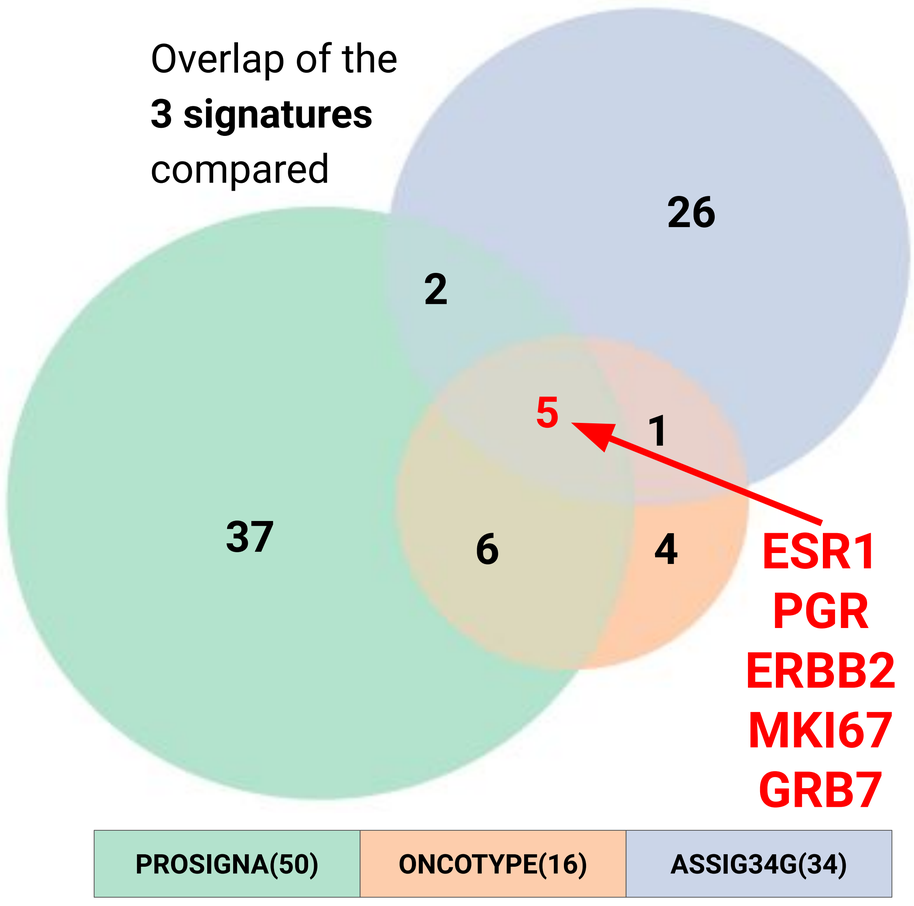
Objective

Discover a robust list of gene expression markers associated with **survival data** and **breast cancer prognosis** that can be biologically interpreted.

Methods

To test and ensure a robust reproducibility, we integrated **two large datasets** (n=**1024** & **879** samples) with genome-wide expression and survival data. Then, we applied an algorithm based on **bagging and the elastic-net classifier** (univariate and multivariate COX regression) to find genes related to IHC markers, prognosis and survival. In this way we found a **survival signature of 34 genes** that was compared with the **Prosigna/PAM50** signature (of 50 genes) and the **Oncotype** signature (of 16 genes). These signatures are used in the **commercial clinical genomic platforms**.

Overlap of the **3 signatures** compared



Lastly, based on the identified **34 genes** we performed an **accurate risk prediction and patient stratification**. This is done by maximizing the log-rank statistic to achieve better separability between the corresponding Kaplan-Meier curves.

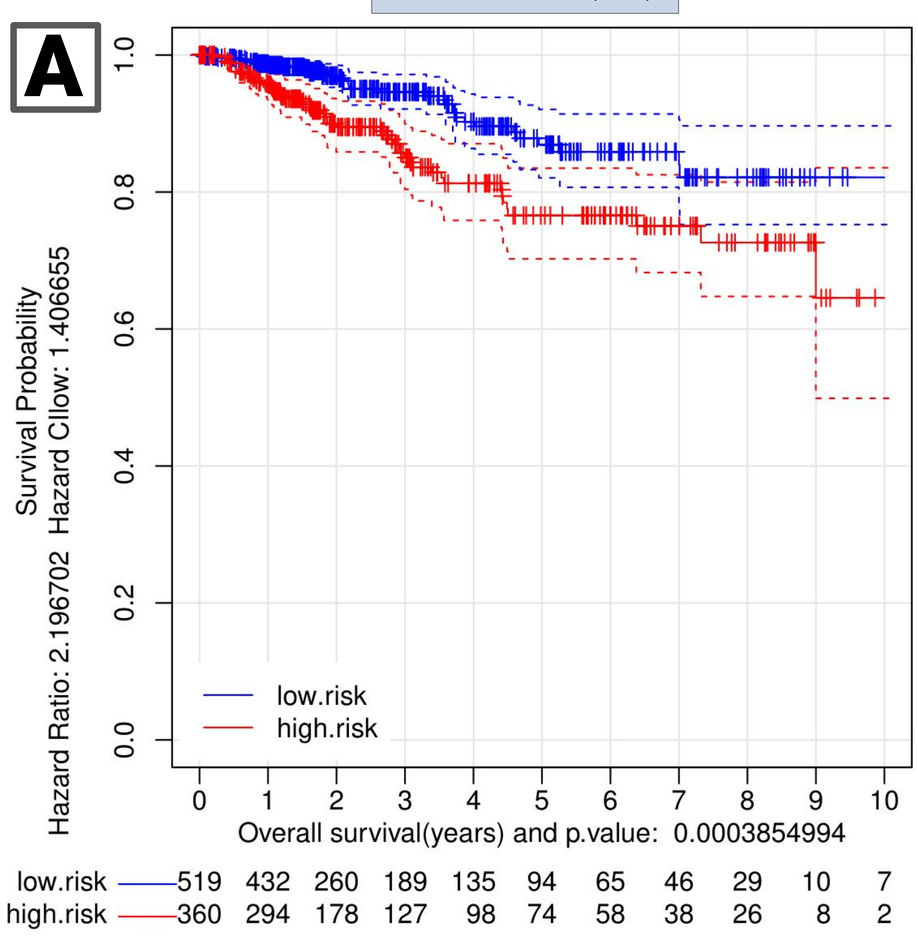
Results

A Kaplan-Meier curves obtained with the **ASSIG34G** signature using the second independent dataset. This set had survival information and genome-wide expression profiles obtained with RNA-seq & included **879** breast tumor samples.

B Breast cancer **patient stratification by risk** (**low risk** & **high risk**) based on the **ASSIG34G** gene signature produced in this study, or based on the signatures used by the genomic platforms **Oncotype** or **Prosigna/PAM50**. Results obtained with the independent set of **879** breast cancer samples.

ASSIG34G(34)

A



B

Signature	logrank p.value	HR	CI.HR
ASSIG34G	0.00038	2.20	(1.41, 3.43)
Oncotype	0.066	1.61	(0.96, 2.69)
Prosigna/PAM50	0.035	1.60	(1.03, 2.49)

- The gene set identified (i.e., the survival signature of 34 genes) **significantly improves** the risk prediction provided by the commercial clinical genomic platforms **Oncotype** and **Prosigna**.
- Furthermore, **some** of the **identified genes have recently been proposed** as **new prognostic markers in breast cancer** and may deserve more attention in current clinical trials to improve **breast tumor risk prediction**.

Acknowledgements

This work is supported by:

- Ministerio de Ciencia e Innovación, Gobierno de España - ISCiii y FEDER (ref. AES 2018, **PI18/00591**).
- Ministerio de Ciencia e Innovación, Gobierno de España - ISCiii y FEDER (ref. IMPaCT-Data, **IMP/00019**).

Reference

Bueno-Fortes, Santiago; Berral-González, Alberto; Sánchez-Santos, José Manuel; Martín Merino, Manuel; De Las Rivas, Javier. **Identification of a gene expression signature for survival and risk prediction of Breast Cancer that improves clinical genomic platforms** (2022) *Bioinformatics Advances*. Submitted.

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